Camera Tracking Research Proof of Concept

OH, RATS!

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1. Overview and Aim

As part of the digital companion for OH, RATS!, I want to investigate whether a camera can detect the position of physical game pieces on the board and communicate this information to the digital application.

The aim is to explore a more seamless connection between the physical board game and its digital companion. Instead of manually scanning a QR code for individual actions, the application could potentially recognise where pieces are located and respond to changes in the game state.

Following a discussion with my supervisor, I will investigate camera based tracking through small poc experiments. The initial focus will be on determining whether the board and its pieces can be detected reliably before attempting to integrate the system into the existing Svelte application.

2. Experiment A: Board Detection and Positioning

Concept

The first experiment will investigate using four unique QR codes or visual markers placed at the corners of a physical board. A camera positioned above the board would detect these markers and use their locations to establish the board's boundaries and orientation.

Once the board has been identified, the software could potentially map the camera image to a grid representing the board's tiles. This would provide a reference system for identifying the positions of game pieces.

Proposed approach

- Place four identifiable markers at the corners of a test area.
- Use a webcam or camera to detect the markers.

- Investigate whether the software can determine the board's boundaries and orientation.
- Explore how the detected area could be divided into a grid of tiles.

For the initial experiment and prototyping, I will use a simple test surface rather than a completed board design.

Technical investigation

I will research computer vision libraries that can detect visual markers and calculate their positions. OpenCV is a potential option, including its ArUco marker detection tools. These markers are designed for computer vision tracking and could be an alternative to ordinary QR codes.

Initial success criteria

- Detect all four corner markers.
- Identify their positions in the camera image.
- Establish a consistent reference area for the board.
- Investigate whether the board's grid can be mapped onto the detected area.

3. Experiment B: Physical Game Piece Tracking

Concept

Once the board's position can be identified, the next experiment will investigate whether individual Cat and Rat pieces can be detected and tracked as they move around the board.

The main challenge is distinguishing the individual pieces and converting their detected camera positions into locations on the board.

For the initial prototype, I will explore attaching unique visual markers to the bases of the pieces. This would allow the software to identify each piece without needing to recognise the full shape of a custom 3D model. If having a full custom shape of a custom 3D model, I will instead use flat tokens for the character pieces (still unique in their design, just more visually accessible for tracking).

Proposed approach

- Create two test markers representing a Cat and a Rat.
- Place them within the area established in Experiment A.
- Use the camera to identify each marker and detect its position.

- Move the markers between different locations and observe whether their positions update.
- Investigate converting their positions into board tiles, such as A1, B2 or C3.

Digital companion integration

If the tracking experiment is successful, I will investigate how to pass the detected piece positions into the existing Svelte application.

This could eventually allow the application to respond to physical gameplay events, such as a Rat entering a trap tile or a Cat reaching a Rat. These interactions would require additional game logic and are not part of the initial tracking test.

Initial success criteria

- Detect two separate game piece markers.
- Distinguish the Cat from the Rat.
- Detect changes in their positions as they move.
- Investigate whether their positions can be matched to individual board tiles.
- Identify limitations, including occlusion/blocking, lighting and tracking accuracy.

4. Tools and Technologies

The following technologies will be investigated during the proof of concept:

- **Camera or webcam:** to capture the board and physical pieces.
- **OpenCV:** a potential computer-vision library for marker detection and position tracking.
- **ArUco markers or QR codes:** identifiable visual markers for locating the board or individual pieces.
- **Python:** a possible environment for the initial tracking experiment.
- **Svelte and JavaScript:** the existing digital companion technology

*** The final choice of tools will depend on the results of the initial experiments** *

5. Limitations and Fallback

Camera based tracking may be affected by lighting, camera angle, obstructed markers, movement and the distance between the camera and the board. Detecting the board does not automatically mean the software can recognise the individual pieces, so these will be tested separately.

The aim is to establish whether camera tracking is feasible within the project's time and technical constraints. If reliable tracking proves too difficult, the existing QR based digital companion will remain the fallback solution.

6. Next Steps

1. Research marker detection and camera based board tracking.
2. Set up a basic camera tracking environment.
3. Test detection of four markers arranged as a board.
4. Test tracking two separate markers representing game pieces.
5. Record screenshots, results, errors and technical limitations.
6. Investigate mapping detected positions to board tiles.
7. If successful, explore connecting the tracking system to the Svelte digital companion.

Reference Links that helped to guide me:

Bing Videos
Hardware description — Smart Chess Board 1.0 documentation
Getting Started With Object Tracking Using OpenCV - GeeksforGeeks
Culturino/Monopoly_tracker: A computer vision pipeline built with OpenCV to detect and track real-time board game events, including optical flow card tracking and dice throw detection.
OpenCV: Detection of ArUco Markers